

Support Geometry and Nearest Neighbours:

Does the Tangential–Orthogonal–Depth Decomposition Add Information
Beyond Feature Lines, Local Hulls and Depth?

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Abstract

For a query point q and its k nearest training neighbours of a class c , the residual $q - \mu$ from the neighbourhood centroid splits into a *tangential* component along the local principal subspace and an *orthogonal* component normal to it, and the position of q inside the neighbourhood can be summarised by a *depth*. The research line SPD001 asked whether this three-part description carries information for classification beyond what nearest feature lines, local hulls and depth measures already use. We make the question precise and answer it. On the theoretical side we show that the regularised local-hyperplane distance of Vincent and Bengio is exactly a data-weighted sum of the orthogonal component and of the squared tangential coordinates, $h_\lambda^2 = O^2 + \sum_j \lambda t_j^2 / (s_j^2 + \lambda)$, interpolating between the hull distance ($\lambda = 0$) and the distance to the local centroid ($\lambda \rightarrow \infty$); the local spatial depth, by contrast, is not a function of the two distances. On the empirical side we run the isolating comparison the line had left pending: 8 datasets (four standard, four synthetic manifolds), 5×5 repeated stratified cross-validation, six reference classifiers (kNN, nearest feature line, HKNN, local PCA hull, local class distance, spatial depth) and a conditional-logit combination of the three components with its one-component-removed ablations, all tuned by leave-one-out accuracy on each training fold, with a criterion fixed before the runs: the decomposition “adds information” only if it beats the best reference on at least 5 of 8 datasets with a paired bootstrap 95% interval above zero. It did so on 0 datasets. Its difference to the best reference ranged from -0.8 to +0.2 accuracy points, and the largest gain compatible with the data at the 95% level is +1.3 points. The ablations locate the information: removing the orthogonal component costs between +0.7 and +5.1 points (interval above zero on 7 of 8 datasets), removing the depth changes accuracy by -0.4 to +0.5 points and removing the tangential component by -0.2 to +0.5 points, in neither case with an interval above zero on any dataset (0 and 0 of 8). A regime scan with off-manifold noise coordinates gives the same picture. The finding answers the question left pending in the line’s public record: the decomposition is a useful bookkeeping device for describing local-hull classifiers, but it does not add predictive information beyond them.

1 Introduction

Nearest-neighbour classification [4, 9] uses only distances from the query to individual training points. A family of extensions replaces the point by a local set built from the query’s neighbours in each class: the line through two prototypes (nearest feature line, 14), the affine hull of the k nearest same-class neighbours with a ridge penalty (HKNN, 23), bounded hulls and hyperdisks [1], or locally adapted metrics [12]. All of them measure, in one way or another, how far the query lies *off* the local class support. A separate tradition, statistical depth [15, 21, 25], measures how central a point is *inside* a cloud, and depth-based classifiers assign the query to the class in which it is deepest [10, 13], with local versions for multimodal classes [17].

The research line SPD001 (“support geometry and nearest neighbours”) set out from the observation that these quantities are components of one decomposition. If μ is the centroid of the k nearest neighbours of q in class c and T_m is the m -dimensional principal subspace of that neighbourhood, then $q - \mu$ splits orthogonally into a tangential part in T_m and an orthogonal part normal to it, and a depth measures where q falls relative to the neighbours. The line’s public record states its objective as *determining whether this decomposition adds information beyond feature lines, hulls and depth measures*, and its findings as *a comparison isolating a specific contribution remains pending; no improvement in predictive accuracy has been validated*. This paper runs that comparison and writes down the answer, together with the elementary facts that explain it.

Contributions. (i) Precise definitions of the tangential, orthogonal and depth components of a query with respect to a class neighbourhood (Section 2). (ii) Three elementary propositions (Section 3): the regularised HKNN distance is an explicit weighted combination of the orthogonal component and the tangential coordinates, monotone in the penalty and bounded by the hull distance and the centroid distance; the nearest-feature-line distance restricted to the same neighbours is sandwiched between the hull distance and the nearest-neighbour distance; the local spatial depth is not a function of the two distances, so it *could* carry information. (iii) The isolating comparison with a criterion stated before the runs (Section 4) and its outcome (Section 5), which is negative under that criterion. (iv) A list of claims with their status (Section 6).

Scope. The paper is a methodological study on small public and synthetic datasets; it makes no claim about any application domain. “Adds information” is given one operational meaning (Section 4); other meanings, such as usefulness for visualisation or for describing existing methods, are not tested and are not denied. Every number is produced by the scripts described in Appendix A with a fixed seed; none is typed by hand.

2 Definitions

Let $(x_i, y_i)_{i=1}^n$ be a training set with $x_i \in \mathbb{R}^d$ and labels $y_i \in \{1, \dots, C\}$, standardised coordinate-wise with training statistics. Fix a query $q \in \mathbb{R}^d$, a class c and an integer k with $k \leq n_c$, where n_c is the number of training points of class c (when q is itself a training point it is excluded from its own neighbourhoods).

Definition 2.1 (class neighbourhood, centred SVD). $N_k^c(q) \subset \{x_i : y_i = c\}$ is the set of the k training points of class c nearest to q in Euclidean distance (ties broken by index). Let $\mu = \mu_c(q)$ be its mean, $V \in \mathbb{R}^{k \times d}$ the matrix with rows $x_i - \mu$, $x_i \in N_k^c(q)$, and $V = W \text{diag}(s_1, \dots, s_p) U^\top$ a thin singular value decomposition with $p = \min(k, d)$, $s_1 \geq \dots \geq s_p \geq 0$ and orthonormal columns $u_1, \dots, u_p$ of U . Write $r = q - \mu$ and $t_j = u_j^\top r$. Since the rows of V sum to zero, $\text{rank } V \leq \min(k - 1, d)$; we call the neighbourhood *generic* when equality holds, and write $\rho = \min(k - 1, d)$.

Definition 2.2 (tangential and orthogonal components). For $1 \leq m \leq \rho$ let $T_m = \text{span}(u_1, \dots, u_m)$ (the local principal, or tangent, subspace of dimension m) and P_m the orthogonal projector onto it. The *tangential distance* and the *orthogonal distance* of q to the class- c neighbourhood are

$$T_m(q, c) = \|P_m r\| = \left(\sum_{j \leq m} t_j^2 \right)^{1/2}, \quad O_m(q, c) = \|(I - P_m)r\| = \left(\|r\|^2 - \sum_{j \leq m} t_j^2 \right)^{1/2}.$$

Definition 2.3 (local hull, HKNN and feature-line distances). The *local hull distance* is $\text{dist}(q, \text{aff } N_k^c(q))$, the Euclidean distance from q to the affine hull of the neighbourhood. For $\lambda \geq 0$ the *penalised local-hyperplane distance* of Vincent and Bengio [23] is

$$h_\lambda(q, c)^2 = \min_{\alpha \in \mathbb{R}^k} \|r - V^\top \alpha\|^2 + \lambda \|\alpha\|^2.$$

The *local feature-line distance* is $\ell(q, c) = \min_{x_i \neq x_j \in N_k^c(q)} \text{dist}(q, x_i + \mathbb{R}(x_j - x_i))$, the nearest-feature-line distance of Li and Lu [14] restricted to the k neighbours. The *local class distance* is $\bar{d}(q, c) = \frac{1}{k} \sum_{x_i \in N_k^c(q)} \|q - x_i\|$.

Definition 2.4 (local spatial depth). With the spatial sign $S(v) = v/\|v\|$ for $v \neq 0$ and $S(0) = 0$, the *local spatial depth* of q with respect to the class- c neighbourhood is

$$D(q, c) = 1 - \left\| \frac{1}{k} \sum_{x_i \in N_k^c(q)} S(x_i - q) \right\| \in [0, 1],$$

the spatial (L_1) depth of Vardi and Zhang [22] and Serfling [19], computed on the neighbourhood instead of on the whole class. We call $E = 1 - D$ the *outlyingness*.

The spatial depth equals 1 exactly at the spatial median of the neighbourhood, which is unique when the neighbours are not collinear, and tends to 0 as $\|q\| \rightarrow \infty$; it is invariant under orthogonal transformations and translations but not under general affine maps [2, 19, 22] [literature]. Restricting a depth to a neighbourhood of the query is a simple proxy for the local depths of Paindaveine and Van Bever [17]; we do not use their symmetrised construction.

Definition 2.5 (classifiers). All classifiers below assign q to the class that minimises a score $\sigma(q, c)$ (or maximises the depth), with no class prior, except kNN, which takes a majority vote among the k nearest neighbours of all classes (ties towards the class of the nearest neighbour).

- NFL: $\sigma = \ell(q, c)$; HKNN: $\sigma = h_\lambda(q, c)$; LPH (local PCA hull): $\sigma = O_m(q, c)$;
- LCD: $\sigma = \bar{d}(q, c)$; SD: $\sigma = E(q, c)$; T: $\sigma = T_m(q, c)$;
- TOD (the decomposition classifier): with the feature vector $f_c(q) = (\log T_m, \log O_m, \log E)(q, c)$, standardised feature-wise with training statistics, and a weight vector $w \in \mathbb{R}^3$ shared by all classes,

$$P(c | q) = \frac{\exp(-w^\top f_c(q))}{\sum_{c'} \exp(-w^\top f_{c'}(q))},$$

a conditional logit without intercepts; w is fitted by maximum likelihood with an L_2 penalty $0.001 \|w\|^2$ on leave-one-out features of the training fold. TO, TD, OD are the same model with one feature removed. (A one-feature model with a positive weight reproduces the corresponding argmin rule, so LPH and SD are the one-component ablations for O and for D .)

The logarithms make the combination a product of powers $T_m^{w_1} O_m^{w_2} E^{w_3}$, invariant to the scale of each feature; a floor of 10^{-6} is added inside each logarithm.

3 Elementary results

Proposition 3.1 (Pythagoras and the hull). *For every $m \leq \rho$, $\|q - \mu\|^2 = T_m^2 + O_m^2$. If the neighbourhood is generic then $O_\rho(q, c) = \text{dist}(q, \text{aff } N_k^c(q)) = h_0(q, c)$.*

Proof. The first identity is orthogonality of P_m and $I - P_m$. For the second, $\text{aff } N_k^c(q) = \mu + \text{row}(V) = \mu + \text{span}(u_j : s_j > 0)$; in the generic case this is $\mu + T_\rho$, so $\text{dist}(q, \text{aff } N_k^c(q)) = \|(I - P_\rho)r\| = O_\rho$. Finally $h_0^2 = \min_\alpha \|r - V^\top \alpha\|^2$ is the squared distance from r to $\text{row}(V)$, the same quantity. $\square$

Proposition 3.2 (HKNN is a weighted tangential–orthogonal combination). *For a generic neighbourhood and every $\lambda > 0$,*

$$h_\lambda(q, c)^2 = O_\rho(q, c)^2 + \sum_{j=1}^{\rho} \frac{\lambda}{s_j^2 + \lambda} t_j^2.$$

Consequently $\lambda \mapsto h_\lambda$ is non-decreasing, $O_\rho \leq h_\lambda \leq \|q - \mu\|$, $h_\lambda \rightarrow O_\rho$ as $\lambda \downarrow 0$ and $h_\lambda \rightarrow \|q - \mu\|$ as $\lambda \rightarrow \infty$.

Proof. Write $V^\top \alpha = U \text{diag}(s) W^\top \alpha$. Decompose $\alpha = W\beta + \gamma$ with $W^\top \gamma = 0$, so that $V^\top \gamma = 0$ and $\|\alpha\|^2 = \|\beta\|^2 + \|\gamma\|^2$, and decompose $r = \sum_j t_j u_j + r_\perp$ with $r_\perp \perp \text{span}(U)$. Then

$$\|r - V^\top \alpha\|^2 + \lambda \|\alpha\|^2 = \sum_{j \leq \rho} (t_j - s_j \beta_j)^2 + \|r_\perp\|^2 + \lambda \|\beta\|^2 + \lambda \|\gamma\|^2.$$

The minimum has $\gamma = 0$ and, for each j , $\beta_j = s_j t_j / (s_j^2 + \lambda)$, with minimal value $(t_j - s_j \beta_j)^2 + \lambda \beta_j^2 = t_j^2 (\lambda^2 + \lambda s_j^2) / (s_j^2 + \lambda)^2 = \lambda t_j^2 / (s_j^2 + \lambda)$. For $j > \rho$ one has $s_j = 0$ in the generic case and the term equals t_j^2 ; together with $\|r_\perp\|^2$ these terms sum to $\|r\|^2 - \sum_{j \leq \rho} t_j^2 = O_\rho^2$ by Proposition 3.1. Each weight $\lambda / (s_j^2 + \lambda)$ lies in $(0, 1]$, increases with λ and tends to 0 or 1 at the two ends, which gives the monotonicity and the limits. $\square$

Remark 3.3. Proposition 3.2 says that HKNN already combines the two distances of Definition 2.2, with an anisotropic weighting of the tangential coordinates that penalises low-variance directions more ($\lambda / (s_j^2 + \lambda)$ is larger when s_j is small). The model TO combines $\log T_m$ and $\log O_m$ with two scalar weights and a cut-off m ; neither family contains the other. The classifier LCD is close to the $\lambda \rightarrow \infty$ end (distance to the local centroid versus mean distance to the neighbours). What the decomposition can add over this family is therefore, in the first place, the depth.

Proposition 3.4 (feature lines). $\text{dist}(q, \text{aff } N_k^c(q)) \leq \ell(q, c) \leq \min_{x_i \in N_k^c(q)} \|q - x_i\|$.

Proof. Every line through two points of $N_k^c(q)$ lies in $\text{aff } N_k^c(q)$, which gives the left inequality; the distance to a line is at most the distance to any of its points, which gives the right one. $\square$

Proposition 3.5 (the depth is not a function of the two distances). *There are neighbourhoods and pairs of queries with equal (T_m, O_m) and different D , and pairs with equal D and different (T_m, O_m) .*

Proof. Take $d = 1$, $N_k^c(q) = \{-1, 0, 3\}$, $k = 3$, so $\mu = 2/3$, $m = \rho = 1$ and $O_1 \equiv 0$. In one dimension $D(q) = 1 - |\frac{1}{k} \sum_i \text{sign}(x_i - q)|$. The queries $q_1 = 8/3$ and $q_2 = -4/3$ have $T_1 = 2$ and $O_1 = 0$, but $D(q_1) = 1 - \frac{1}{3} = \frac{2}{3}$ (two neighbours below, one above) and $D(q_2) = 1 - 1 = 0$ (all three above). The queries $q_3 = 1$ and $q_4 = 5/3$ both have $D = \frac{2}{3}$ but $T_1 = \frac{1}{3}$ and $T_1 = 1$ respectively. $\square$

Propositions 3.1–3.5 are elementary [proved here]. They fix expectations: relative to HKNN and LPH, the only component that is not already in play is the depth, and Proposition 3.5 shows that it is formally free to help. Whether it does is the empirical question.

4 Protocol and predefined criterion

Datasets. Four standard sets from scikit-learn [18]: iris [8] ($n = 150$, $d = 4$, $C = 3$), wine [6] (178, 13, 3), Wisconsin diagnostic breast cancer [20] (569, 30, 2) and a stratified subsample of the digits (1000 images, 100 per class, $d = 64$). Four synthetic manifolds with $n = 600$: a swiss roll (`make_swiss_roll`, noise 1.0) with three classes given by the terciles of the roll parameter, so that class boundaries run *along* the manifold; two moons (noise 0.3) in the plane; the same moons with ten independent $N(0, 1)$ coordinates appended ($d = 12$), the off-manifold-noise regime; and two nested spheres in $\mathbb{R}^3$ (radii 1 and 1.5, 300 points each, isotropic Gaussian noise of standard deviation 0.2). The digits subsample, the noise coordinates and the spheres are drawn once with the master seed.

Evaluation. 5×5 repeated stratified k -fold cross-validation (25 folds, identical for all methods). Within each fold, coordinates are standardised with training statistics. Every hyper-parameter is chosen on the training fold by leave-one-out accuracy over a fixed grid, ties going to the first grid point: $k \in \{1, 3, 5, 7, 9, 11, 15, 21, 31, 41\}$ for kNN; $k \in \{5, 10, 20, 30\}$ for all class-neighbourhood methods; $m \in \{1, 2, 3, 5, 8\}$ restricted to $m \leq \min(k - 1, d - 1)$ for LPH, T and the logit models; $\lambda = \lambda_{\text{rel}} \bar{s}^2$ with $\lambda_{\text{rel}} \in \{0.0, 0.1, 1.0, 10.0, 100.0\}$ for HKNN, where $\bar{s}^2$ is the median over training points of the mean squared singular value of their own-class neighbourhood. Leave-one-out features of the training points (each point excluded from its own class neighbourhood) also serve to fit the logit weights. The test metrics are accuracy and *selective accuracy at 80% coverage* [3]: accuracy on the 80% of test points with the highest confidence, the confidence being the vote fraction for kNN, the relative margin $(\sigma_{(2)} - \sigma_{(1)})/(\sigma_{(1)} + \sigma_{(2)})$ between the two best scores for the argmin classifiers and the maximal posterior for the logit models.

Statistics. For two methods, the paired fold-level differences in accuracy ($J = 25$) are summarised by their mean and a percentile bootstrap 95% interval with 10 000 resamples [7], seeded. Folds of repeated cross-validation are not independent, so this interval is somewhat too narrow [5]; as a check we also report the two-sided p -value of the corrected resampled t -test of Nadeau and Bengio [16], with variance inflated by $1/J + n_{\text{test}}/n_{\text{train}} = 1/J + 1/4$. The predefined criterion uses the bootstrap interval, which is the *less* demanding of the two for a positive result.

Criterion (fixed before the runs). For each dataset the *best reference* is the one of kNN, NFL, HKNN, LPH, LCD, SD with the highest mean accuracy. **C1**: the decomposition “adds information” if and only if TOD beats the best reference on at least 5 of the 8 datasets with a bootstrap interval strictly above zero. Choosing the best reference after the fact biases the comparison *against* TOD (winner’s curse on the reference), so a positive C1 would have been strong evidence; a negative C1 is complemented by the comparison against each reference separately. **C2** (attribution): the same rule applied to TOD versus each ablation identifies which component carries any gain. **C3** (secondary): the same rule on selective accuracy. **E2** (regime): the moons with $p \in \{0, 2, 5, 10, 20\}$ appended noise coordinates, to look for a regime of low-dimensional support with off-manifold noise in which some component might help.

Note on the grids. A first full run used $k \leq 20$ for the neighbourhood methods, $k \leq 21$ for kNN and $\lambda_{\text{rel}} \leq 10$; the chosen values sat at the upper end of these grids on several datasets, so the grids were widened to the values above and the whole experiment was rerun. The criterion and everything else were unchanged; both runs are kept with the results and reach the same verdict.

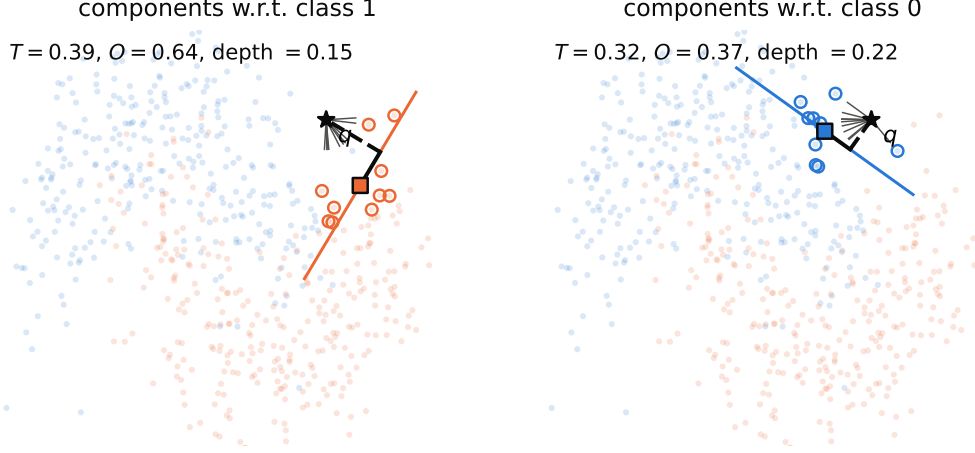


Figure 1: The decomposition on the two-moons data for one query q (star) with respect to each class: circles are the $k = 10$ class neighbours, the square their centroid, the line the one-dimensional tangent subspace T_1 ; the solid segment is the tangential component, the dashed segment the orthogonal one; the short rays are the unit vectors $S(x_i - q)$ whose mean defines the spatial depth.

dataset	n	d	C	references						decomposition and ablations				
				kNN	NFL	HKNN	LPH	LCD	SD	TOD	TO	TD	OD	T
iris	150	4	3	94.9	95.2	95.7	96.4	95.7	93.9	96.1	95.6	93.6	96.3	93.6
wine	178	13	3	96.3	97.0	97.0	95.9	96.9	96.4	97.1	96.9	96.4	97.3	89.5
breast cancer	569	30	2	96.6	96.0	97.0	96.7	96.9	95.5	96.7	97.1	95.7	96.7	93.3
digits	1000	64	10	96.4	97.6	97.4	97.4	96.9	91.0	97.2	97.3	92.1	97.1	43.6
swiss roll	600	3	3	96.8	92.9	97.4	97.0	96.9	95.4	97.2	97.2	95.2	97.1	78.9
moons	600	2	2	89.9	80.0	89.9	88.9	89.9	88.6	89.1	89.1	88.0	89.0	78.7
moons+10 noise	600	12	2	82.8	77.7	82.8	82.1	82.8	80.0	82.5	82.6	79.8	81.9	62.8
spheres	600	3	2	82.5	78.6	85.5	84.8	83.5	81.9	85.7	85.2	82.2	85.8	70.8

Table 1: Mean test accuracy (%) over the 25 folds. The best reference on each dataset is in bold. TOD = tangential + orthogonal + depth; TO, TD, OD drop one component; T, LPH (=O) and SD (=D) are the single components.

5 Results

Table 1 gives the mean cross-validated accuracy of every method, Table 2 the primary comparison, Table 3 the comparison against each reference and Table 4 the ablations; Figure 2 shows the last two graphically. The whole experiment, including the regime scan, took 294 CPU-seconds.

dataset	best ref.	its acc.	TOD acc.	TOD – best ref. [95% CI]	W/L/T	NB p
iris	LPH	96.40	96.13	−0.27 [−1.20, +0.67]	4/5/16	0.834
wine	NFL	96.96	97.07	+0.10 [−1.03, +1.33]	8/9/8	0.950
breast cancer	HKNN	96.98	96.73	−0.25 [−0.70, +0.21]	8/11/6	0.700
digits	NFL	97.56	97.22	−0.34 [−0.64, −0.06]	5/13/7	0.414
swiss roll	HKNN	97.37	97.17	−0.20 [−0.60, +0.17]	7/9/9	0.721
moons	HKNN	89.90	89.13	−0.77 [−1.27, −0.27]	5/16/4	0.290
moons+10 noise	HKNN	82.83	82.47	−0.37 [−1.30, +0.57]	10/11/4	0.779
spheres	HKNN	85.47	85.70	+0.23 [−0.53, +0.97]	11/10/4	0.824

Table 2: Primary comparison (C1). Accuracies in %, differences in accuracy points; W/L/T counts folds won, lost and tied by TOD; NB p is the corrected resampled t -test. Significant wins under C1: 0 of 8 (5 required); significant losses: 2 (digits, moons).

dataset	vs kNN	vs NFL	vs HKNN	vs LPH	vs LCD	vs SD
iris	+1.2 [+0.0, +2.4]	+0.9 [-0.1, +1.9]	+0.4 [-0.4, +1.3]	-0.3 [-1.2, +0.7]	+0.4 [-0.7, +1.6]	+2.3 [+1.1, +3.5]
wine	+0.8 [-0.5, +2.0]	+0.1 [-1.1, +1.3]	+0.1 [-1.0, +1.3]	+1.1 [-0.1, +2.4]	+0.1 [-0.8, +1.0]	+0.7 [-0.6, +2.1]
breast cancer	+0.1 [-0.4, +0.6]	+0.7 [+0.1, +1.3]	-0.2 [-0.7, +0.2]	+0.0 [-0.5, +0.5]	-0.1 [-0.6, +0.3]	+1.3 [+0.7, +1.9]
digits	+0.8 [+0.4, +1.2]	-0.3 [-0.6, -0.1]	-0.1 [-0.4, +0.1]	-0.2 [-0.4, +0.1]	+0.4 [+0.1, +0.7]	+6.2 [+5.5, +6.8]
swiss roll	+0.3 [-0.2, +0.9]	+4.3 [+3.3, +5.3]	-0.2 [-0.6, +0.2]	+0.2 [-0.2, +0.6]	+0.3 [-0.2, +0.8]	+1.8 [+1.0, +2.6]
moons	-0.8 [-1.6, +0.2]	+9.2 [+7.2, +11.1]	-0.8 [-1.3, -0.2]	+0.2 [-0.3, +0.8]	-0.8 [-1.4, -0.1]	+0.5 [-0.2, +1.3]
moons+10 noise	-0.3 [-1.2, +0.5]	+4.8 [+3.6, +6.0]	-0.4 [-1.3, +0.5]	+0.3 [-0.6, +1.2]	-0.4 [-1.3, +0.6]	+2.5 [+1.3, +3.6]
spheres	+3.2 [+1.8, +4.5]	+7.1 [+6.1, +8.1]	+0.2 [-0.5, +1.0]	+0.9 [+0.2, +1.5]	+2.2 [+0.9, +3.5]	+3.8 [+2.6, +4.9]

Table 3: TOD minus each reference, in accuracy points, mean and bootstrap 95% interval over folds.

dataset	– depth (vs TO)	– orthogonal (vs TD)	– tangential (vs OD)	vs LPH (O only)	vs SD (D only)
iris	+0.5 [-0.4, +1.5]	+2.5 [+1.3, +3.7]	-0.1 [-1.2, +0.8]	-0.3 [-1.2, +0.7]	+2.3 [+1.1, +3.5]
wine	+0.2 [-1.0, +1.5]	+0.7 [-0.6, +2.0]	-0.2 [-1.1, +0.7]	+1.1 [-0.1, +2.4]	+0.7 [-0.6, +2.1]
breast cancer	-0.4 [-0.7, -0.0]	+1.1 [+0.5, +1.7]	+0.0 [-0.5, +0.4]	+0.0 [-0.5, +0.5]	+1.3 [+0.7, +1.9]
digits	-0.1 [-0.3, +0.1]	+5.1 [+4.4, +5.8]	+0.1 [-0.1, +0.2]	-0.2 [-0.4, +0.1]	+6.2 [+5.5, +6.8]
swiss roll	-0.1 [-0.4, +0.3]	+1.9 [+1.2, +2.7]	+0.1 [-0.4, +0.5]	+0.2 [-0.2, +0.6]	+1.8 [+1.0, +2.6]
moons	+0.0 [-0.4, +0.4]	+1.1 [+0.2, +2.0]	+0.1 [-0.3, +0.6]	+0.2 [-0.3, +0.8]	+0.5 [-0.2, +1.3]
moons+10 noise	-0.2 [-1.1, +0.7]	+2.7 [+1.5, +3.8]	+0.5 [-0.4, +1.5]	+0.3 [-0.6, +1.2]	+2.5 [+1.3, +3.6]
spheres	+0.5 [-0.0, +1.0]	+3.5 [+2.3, +4.7]	-0.1 [-0.9, +0.7]	+0.9 [+0.2, +1.5]	+3.8 [+2.6, +4.9]

Table 4: Ablations (C2): TOD minus the model with the named component removed, in accuracy points, mean and bootstrap 95% interval. A positive value means the component helped.

C1: the decomposition against the best reference. The best reference was a hull method (HKNN, LPH or NFL) on 8 of the 8 datasets (Table 2). TOD beat it with an interval above zero on 0 of 8 datasets, against 5 required; the criterion is not met. The mean difference ranged from -0.8 to +0.2 points and was within one point on 8 of 8 datasets; its bootstrap interval lay entirely below zero on 2 datasets (digits, moons): -0.34 points on digits [-0.6, -0.1] and -0.77 on moons [-1.3, -0.3]. Neither loss survives the corrected t -test ($p = 0.414$ and $p = 0.290$), which is the cautious reading in both directions: the data neither show a gain nor establish a loss beyond a fraction of a point. The largest upper interval limit over all datasets is +1.3 points: gains larger than that are excluded at the 95% level everywhere; smaller gains are not excluded by this study.

Against each reference. Against HKNN, the reference closest in spirit, the difference ranged from -0.8 to +0.4 points, with 0 intervals above zero and 1 below (Table 3); against LPH from -0.3 to +1.1 (1 above zero). The one dataset on which TOD beats the plain PCA hull with an interval above zero is the spheres (+0.9 points), where the depth is the component that helps (TOD – TO = +0.5 [-0.0, +1.0]); but HKNN closes the same gap with its penalty, and TOD – HKNN is +0.2 points there. TOD beat kNN with an interval above zero on 2 datasets and below zero on 0, and beat the depth-only classifier SD on 6 of 8; the hull classifiers do the same, so these are the known advantages of local hulls over points and over depth [23], not a contribution of the decomposition. The accuracies of HKNN, LPH, TOD, TO and OD never differ by more than 1.3 points on any dataset: they behave as one family, as Proposition 3.2 and Remark 3.3 suggest.

C2: where the information is. Removing the orthogonal component costs between +0.7 and +5.1 points, with an interval above zero on 7 of 8 datasets (Table 4). Removing the depth changes accuracy by -0.4 to +0.5 points, with 0 intervals above zero and 1 below zero (breast cancer, upper limit -0.0, i.e. marginal); removing the tangential component changes it by -0.2 to +0.5 points, with 0 intervals above zero and 0 below. The largest upper limits for the depth and tangential contributions are +1.5 and +1.5 points. The fitted weights tell the same story (Table 5): the weight on $\log O$ lies between 1.52 and 3.93 across datasets, the weight on $\log T$ between 0.17 and 1.38, and the weight on $\log E$ between -1.15 and 1.81, negative on 3 datasets, so its sign is not even stable. The tangential distance alone is a poor classifier (T: from 43.6% on

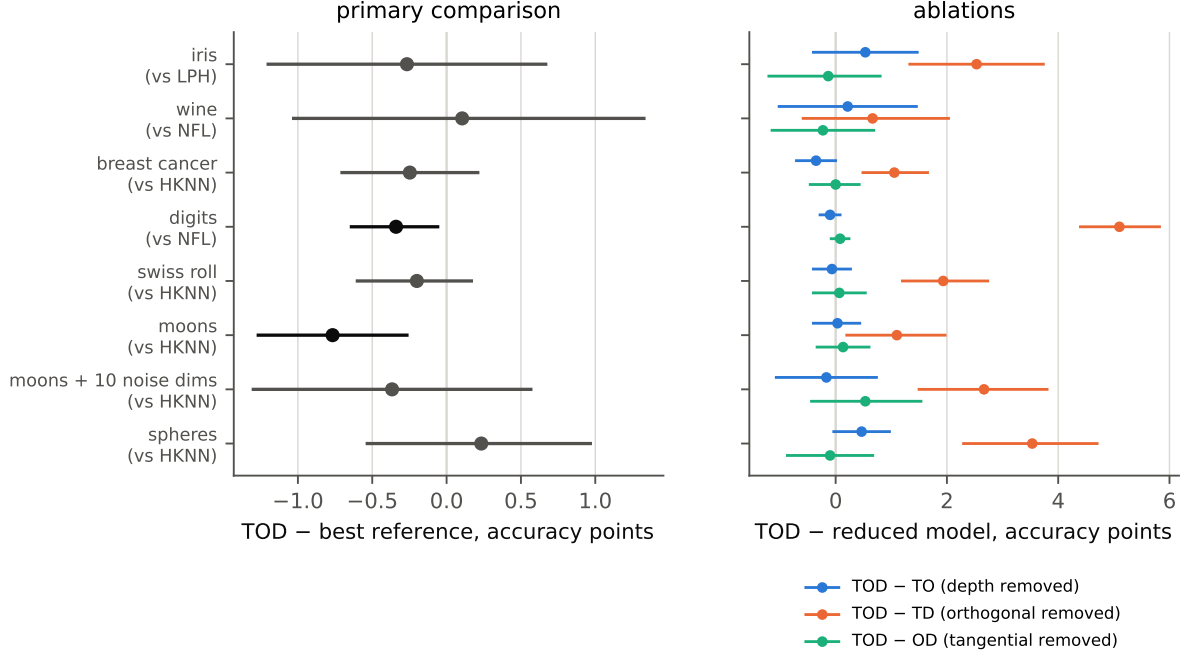


Figure 2: Left: TOD minus the best reference per dataset (C1), mean and bootstrap 95% interval; black marks an interval excluding zero. Right: the three ablations (C2).

dataset	kNN k	HKNN (k, λ_{rel})	LPH (k, m)	SD k	TOD (k, m)	w_T	w_O	w_D
iris	11	(30, 0.1)	(20, 2)	10	(20, 2)	0.82	2.95	1.40
wine	31	(30, 1)	(20, 2)	20	(30, 8)	1.38	3.09	1.81
breast cancer	3	(20, 10)	(20, 1)	20	(20, 1)	0.71	2.91	0.89
digits	1	(20, 0.1)	(10, 8)	10	(10, 8)	0.29	3.93	0.52
swiss roll	5	(5, 1)	(5, 1)	10	(10, 1)	0.17	3.30	0.57
moons	41	(30, 10)	(20, 1)	30	(30, 1)	0.42	1.52	-0.12
moons+10 noise	41	(30, 100)	(30, 1)	30	(30, 5)	0.37	3.04	-1.15
spheres	7	(5, 10)	(5, 1)	10	(10, 1)	0.26	1.92	-0.53

Table 5: Modal hyper-parameters over the 25 folds and mean fitted weights of TOD on the standardised features $\log T_m$, $\log O_m$, $\log E$ (larger weight = more influence; positive = smaller value favours the class).

digits to 93.6%), consistent with it being a within-support coordinate rather than an off-support one.

C3: abstention. With each method abstaining on its own least confident fifth of the test points, TOD differed from the best reference by -1.0 to +0.2 points; the interval lay above zero

dataset	kNN	NFL	HKNN	LPH	LCD	SD	TOD	TO	OD	TOD – best ref.	[CI]
iris	99.0	99.5	99.8	99.7	99.3	97.8	99.3	99.3	98.8	-0.3	[-0.8, +0.0]
wine	99.2	100.0	100.0	99.7	99.7	98.7	99.7	100.0	99.4	-0.3	[-0.7, +0.0]
breast cancer	98.9	98.8	99.3	99.1	99.3	98.4	99.4	99.3	99.4	+0.0	[-0.2, +0.3]
digits	99.3	99.5	99.6	99.5	99.6	98.1	99.7	99.6	99.5	+0.2	[+0.0, +0.3]
swiss roll	99.5	98.0	99.6	99.6	99.7	99.0	99.5	99.5	99.5	-0.1	[-0.2, +0.1]
moons	95.2	85.1	93.8	92.8	96.0	91.4	92.8	92.8	92.9	-1.0	[-1.6, -0.5]
moons+10 noise	88.3	82.2	88.5	87.7	88.9	85.9	87.5	87.5	87.8	-1.0	[-1.7, -0.3]
spheres	88.0	83.3	90.9	89.9	89.1	86.4	90.8	90.0	90.5	-0.1	[-0.8, +0.5]

Table 6: Selective accuracy (%) at 80% coverage (C3), each method abstaining on its own least confident 20% of test points; last column against the best reference of Table 2.

p	kNN	NFL	HKNN	LPH	LCD	SD	TOD	TO	TD	OD	best ref.	TOD – best ref.	[CI]
0	89.9	80.0	89.9	88.9	89.9	88.6	89.1	89.1	88.0	89.0	HKNN	-0.8	[-1.3, -0.3]
2	87.1	82.6	86.4	86.9	87.9	85.5	87.8	86.9	85.5	87.5	LCD	-0.1	[-0.7, +0.6]
5	84.4	81.9	83.9	83.9	84.9	82.3	84.6	84.0	82.5	84.3	LCD	-0.3	[-0.9, +0.4]
10	82.8	77.7	82.8	82.1	82.8	80.0	82.5	82.6	79.8	81.9	HKNN	-0.4	[-1.3, +0.5]
20	80.9	75.7	81.4	80.6	81.3	79.1	81.0	81.1	78.7	80.8	HKNN	-0.4	[-1.2, +0.3]

Table 7: E2, regime scan: mean accuracy (%) on the two moons (noise 0.3, $n = 600$) with p appended $N(0, 1)$ coordinates, same protocol as Table 1.

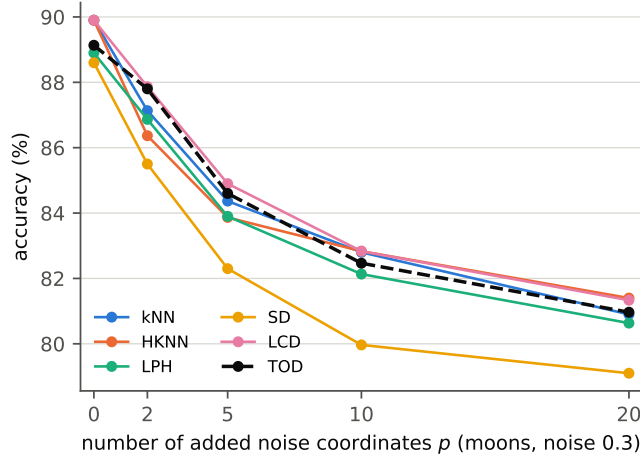


Figure 3: E2: accuracy against the number p of appended noise coordinates for five references and TOD (dashed).

on 1 dataset (digits, +0.2 points) and below zero on 2 (moons and moons with noise, -1.0 and -1.0) (Table 6). The posterior of the three-component logit is therefore not a better abstention signal than the margins of the reference scores; in particular the depth, which one might expect to flag off-support queries, does not improve selective accuracy here.

E2: the off-manifold-noise regime. Appending noise coordinates to the moons lowers every method (Table 7, Figure 3): kNN by 9.0 points from $p = 0$ to $p = 20$, HKNN by 8.5, TOD by 8.2, SD by 9.5. The best reference at each p was one of {HKNN, LCD}; TOD minus the best reference ranged from -0.8 to -0.1 points with 0 intervals above zero and 1 below. The depth contribution (TOD – TO) ranged from -0.2 to +0.9 points and the tangential contribution (TOD – OD) from +0.1 to +0.5, with 1 and 0 intervals above zero respectively; the orthogonal contribution (TOD – TD) ranged from +1.1 to +2.7. The single interval above zero for the depth contribution in the whole study occurs at $p = 2$ (+0.9 points [+0.3, +1.5]), and there TOD remains -0.1 points from the best reference (LCD, 87.9%): with a few noise coordinates the depth helps the logit model catch up with the references, not overtake them. There is thus no noise regime, in this range, in which the depth or the tangential component adds to what the best reference already achieves.

6 What can be claimed

Table 8 lists the claims with their status. The answer to the line’s question, in the operational sense fixed in Section 4, is negative: the tangential–orthogonal–depth decomposition does not add predictive information beyond the local hull. Proposition 3.2 says why the tangential part cannot be expected to: the regularised hull distance already contains it with a sensible weighting. The depth was formally free to help (Proposition 3.5) and did not add to the best reference,

claim	status
$\ q - \mu\ ^2 = T_m^2 + O_\rho^2$; O_ρ is the local hull distance h_0 (Prop. 3.1)	proved here
$h_\lambda^2 = O_\rho^2 + \sum_j \lambda t_j^2 / (s_j^2 + \lambda)$; h_λ monotone in λ between O_ρ and $\ q - \mu\ $ (Prop. 3.2)	proved here
Local feature-line distance lies between hull distance and nearest-neighbour distance (Prop. 3.4)	proved here
Local spatial depth is not a function of (T_m, O_m) , nor conversely (Prop. 3.5)	proved here
Spatial depth: range $[0, 1]$, maximal at the spatial median, orthogonally invariant	literature
C1: TOD beats the best reference on 0 of 8 datasets (5 required); the decomposition does not add information in the predefined sense	verified (negative)
Upper 95% limit of the gain over the best reference is at most +1.3 points on every dataset	verified
C2: the orthogonal component carries the information (interval above zero on 7/8); depth and tangential components removable without significant loss (0/8 and 0/8)	verified
C3: no gain in selective accuracy at 80% coverage (1 above zero, 2 below)	verified
E2: no off-manifold-noise regime with $p \leq 20$ in which TOD exceeds the best reference; the depth helps the logit model only at $p = 2$ (+0.9 points), without overtaking the references	verified
HKNN, LPH, TOD, TO, OD differ by at most 1.3 points on every dataset	verified
Local hulls beat kNN on some datasets and beat depth-only classification on most	literature / verified
Gains smaller than about one accuracy point; behaviour on large or high-dimensional data; other depths	not established

Table 8: Claims and their status. “Verified” means produced by the scripts of Appendix A with the fixed seed on the datasets of Section 4; it is not a general statement.

in accuracy or in abstention, on any of the 8 datasets or in the noise regime scanned; its one significant contribution ($p = 2$ in E2) brought the logit model level with the references, not past them.

Limitations. The datasets are small and, for four of them, close to the accuracy ceiling, so the study has power to exclude gains of about one point and not smaller ones; the fold-level bootstrap is anti-conservative and the Nadeau–Bengio p -values, all reported, are the more cautious reading. Hyper-parameters were chosen by leave-one-out accuracy on the training fold and the logit weights were fitted on the same leave-one-out features, an optimism of three parameters shared by all logit models; the references have one or two tuned parameters each. Only one depth (local spatial depth) and one combination rule (a conditional logit without intercepts) were tried; a different depth, per-class weights or a non-linear combiner could behave differently, and the negative finding is about the tested instance, not about every conceivable use of the components. The tangent dimension m was tuned globally, not per neighbourhood by explained variance. No class priors are used by the score-based methods, which handicaps them slightly on the imbalanced breast-cancer data relative to kNN.

Next steps for the research line. The line’s stated objective is discharged with a negative answer and can be closed. If it were reopened, the only directions the present results leave open are: (a) depths with a different mechanism (halfspace or simplicial depth on the neighbourhood, or the symmetrised local depth of 17) tested under the same criterion; (b) the use of the decomposition as a *descriptive* tool for local-hull methods, e.g. choosing λ from the spectrum (s_j) via Proposition 3.2 rather than by search, which is a question about HKNN and not about a new classifier; (c) larger and higher-dimensional data, where the hull methods differ more from kNN, with the caveat that the study would then be about the hull family and not about the

added components.

A Reproducibility

The directory contains `experiments/support_geometry.py` (the whole experiment, E1 and E2; flag `-fast` for a reduced run), `experiments/make_figures.py` and `experiments/make_numbers.py`, which converts `results/results.json` and `results/results_regime.json` into the macros and table bodies used by this document, so that every number in the text is generated and none is typed. The run used seed 20260930, Python 3.11.15, NumPy 2.4.6 [11], SciPy 1.17.1 [24] and scikit-learn 1.9.1 [18], single-threaded BLAS, and took 310 s wall / 294 s CPU on a shared four-core machine. The first run with the narrower grids is kept as `results/tables_run1_narrow_grids.md`.

Computation. For each fold, the squared distances from all queries (training points for leave-one-out, test points for evaluation) to all training points are computed once; for each class and each k in the grid the k nearest same-class points are found by partial sorting, the centred neighbourhood is decomposed by a batched thin SVD, and t_j^2 , s_j^2 , $\|r\|^2$, the feature-line distance (from the Gram matrix of $q - x_i$), the mean neighbour distance and the spatial depth are stored. All scores of Definitions 2.2–2.4 and the closed form of Proposition 3.2 are then evaluated from these arrays; the closed forms were checked against brute-force minimisation and against `lstsq` before the runs. Logit weights are fitted by L-BFGS-B from $w = (1, 1, 1)$ with the analytic gradient. Bootstrap resamples use a generator seeded with 20260930; the corrected t uses $J - 1$ degrees of freedom.

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